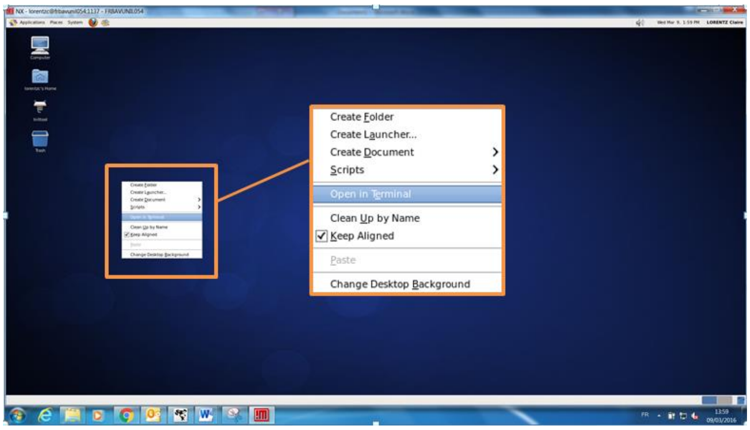
PROCEDURE OF IDENTIFICATION OF MATERIAL PARAMETERS WITH Z-SET SOFTWARE

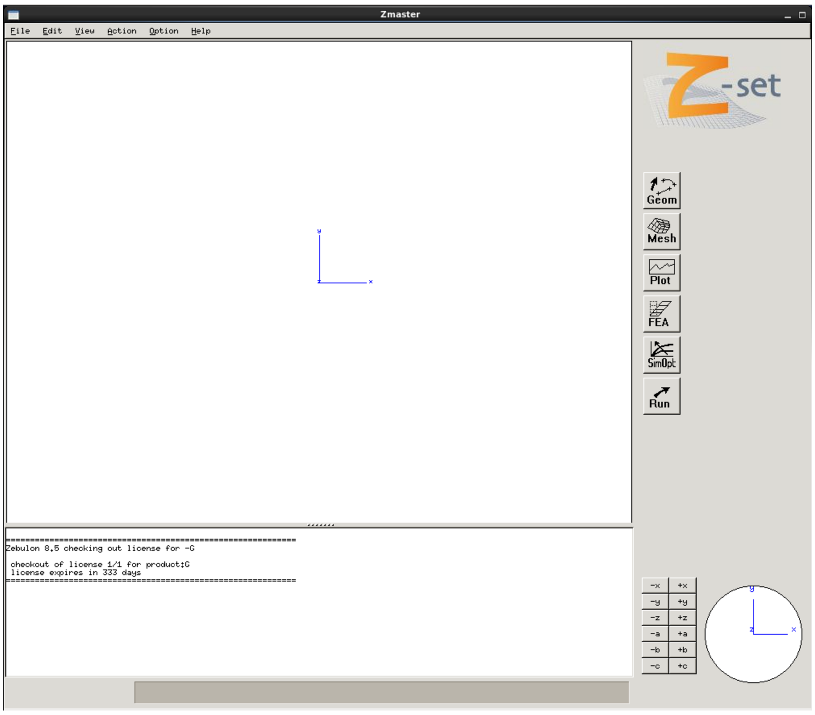
EPNL KINEMATIC PROCEDURE

I. HOW TO START Z-SET SOFTWARE

Right-click anywhere on this window and choose 'Open in Terminal' option as shown after:



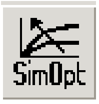
1. A terminal window is now open; you need to type the following command: "Zopt". 'Zopt' opens Z-set software interface.
2. Z-set is now ready to be use.



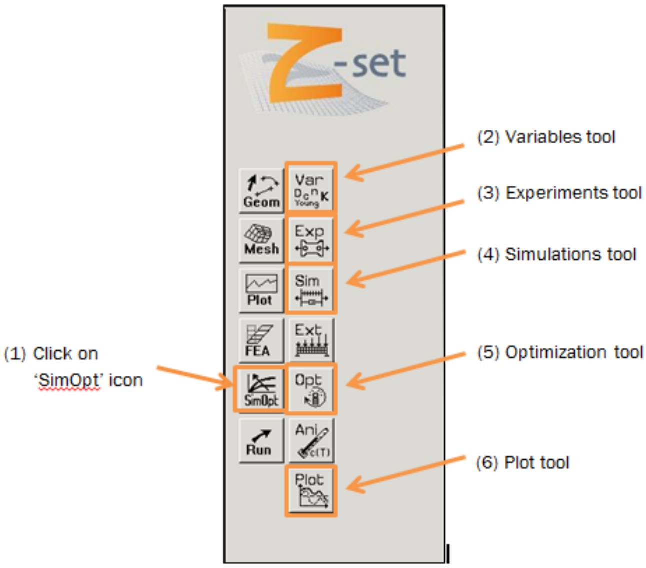
II. How to use Z-set to identify the material parameters from a single-step test

1. Interface

On Z-set interface, we can see six icons but we are going to use only one of them:

SimOpt 

. When you click on this icon, seven tools appear as shown next and we will use them from top to bottom.

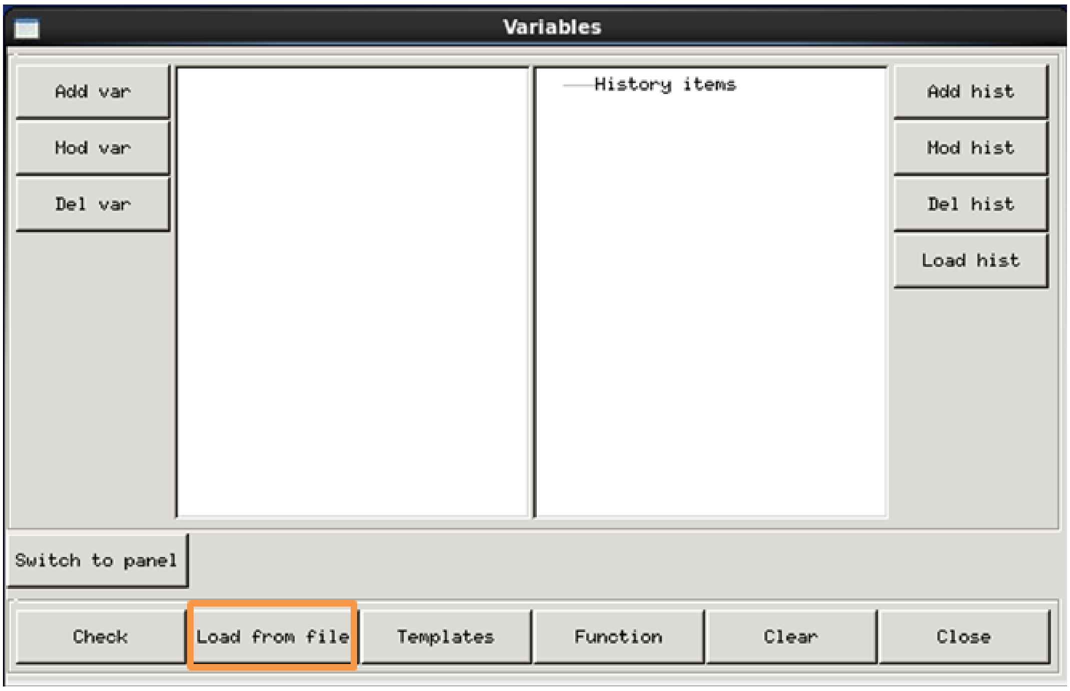


2. Variables tool

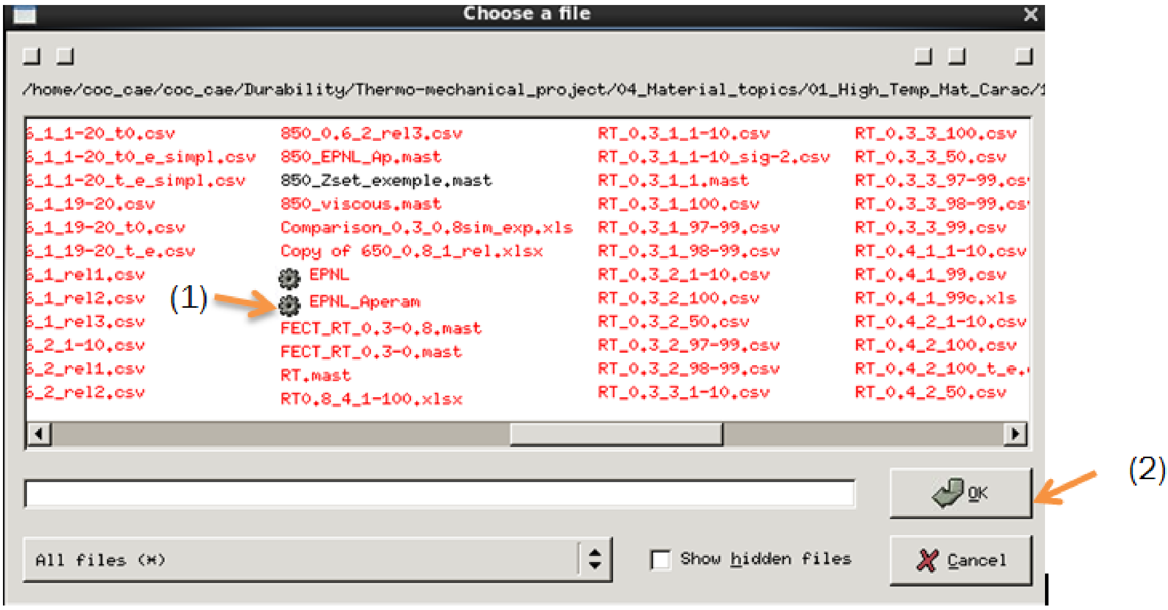
When you click on Variables tool, a new window is open and you define the Chaboche model you use to identify the material parameters by loading the file call 'EPNL_Aperam'. In the 'Variables' window we are going to click on "Load from file" tab.

The EPNL law is using five different parameters:

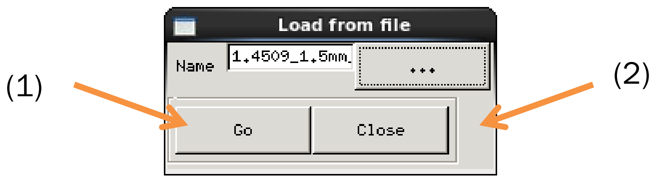
- Elasticity's modulus E
- Poisson coefficient v
- Elasticity domain R_0
- Kinematic hardening parameters C and D



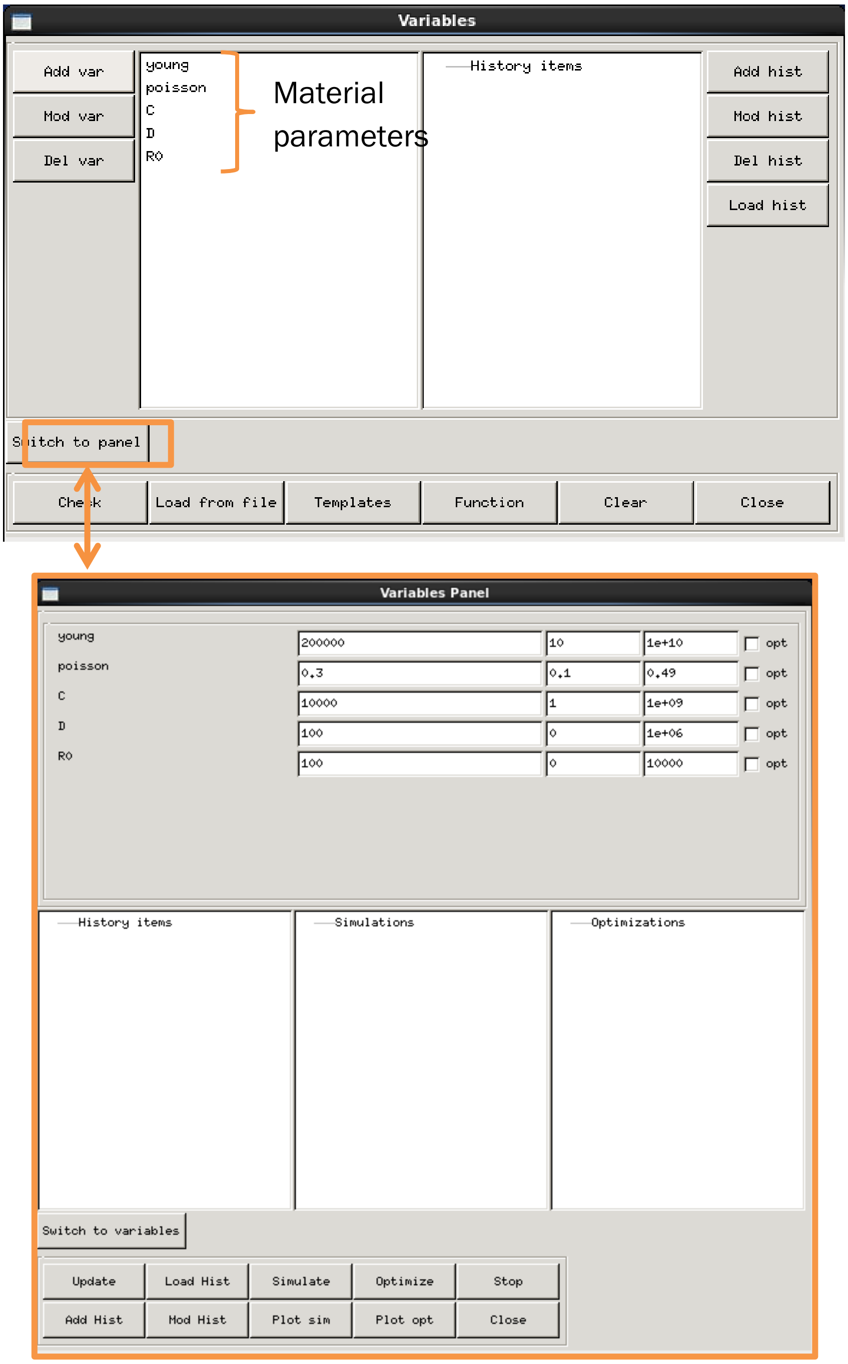
Then you have to find the file call ‘EPNL_Aperam’ next to a gear logo. The path to find the file is: */home/coc_cae/coc_cae/Durability/Thermo-mechanical_project/04_Material-topics/01_High_Temp_Mat_Carac/1.4509_1.5mm_Thyssen/1.4509Thyssen_Zset*



Once you click on OK, you should click Go to download the variables on the tool and then click Close.



Now we can see the five parameters we want to identify with this model: E, v, R0, C and D in the Variables window and if we want to change their values, you have to click on Switch to panel.

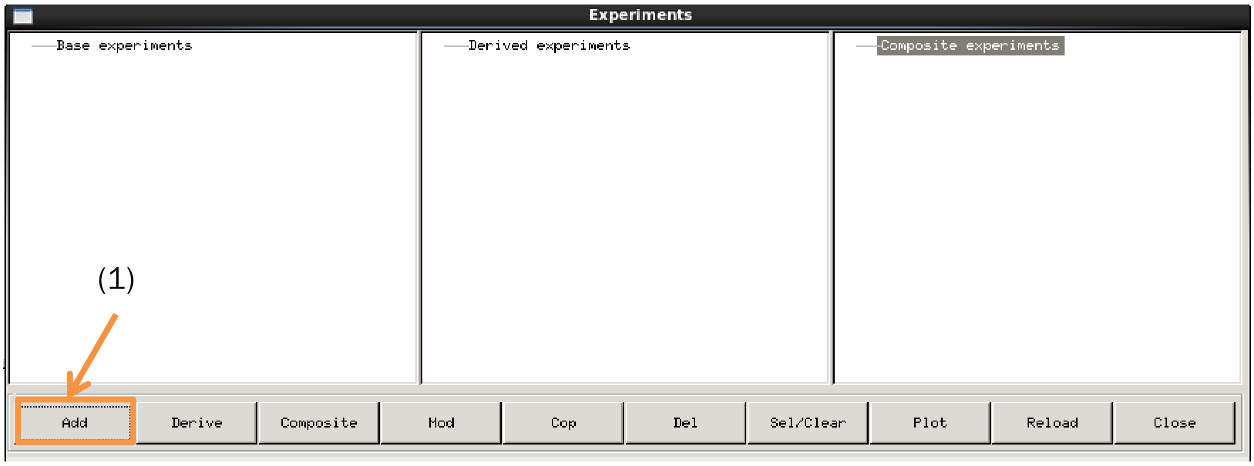


3. EXPERIMENTS TOOL

The experiment tool is use to generate plots from experimental results and to filter those results to characterize the material we study. There are two types of experiments: single step test and multiple steps test. I will detail the procedure for the single step test.

a. Add an experiment

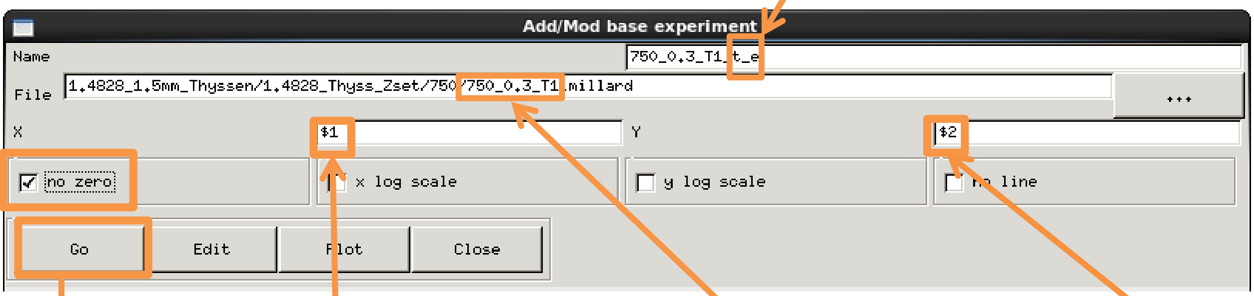
When you click on ‘Exp’ icon, the next window is opening and the first thing to do is to click on ‘Add’ to add the first plot from our experimental results:



When this window is open, you have to load the file with your experimental results. This file is composed of numerical data structured into three columns, first one is the time, second one is strain ‘eto’ ε and third one is stress ‘sigma’ σ. We have to plot three curves with every result: t-e, t-s, e-s, to do that you need to change the number in the X and Y section. Don’t forget to tick the ‘No zero’ option if the first time of your result is not zero and name this experiment carefully; see the example below.

(2) We want to plot the time and the strain (t and e)

(4)



(5)

(3) X axis: time, first column

(1) Example test: Test 1 at 750°C with an axial strain amplitude of 0.3

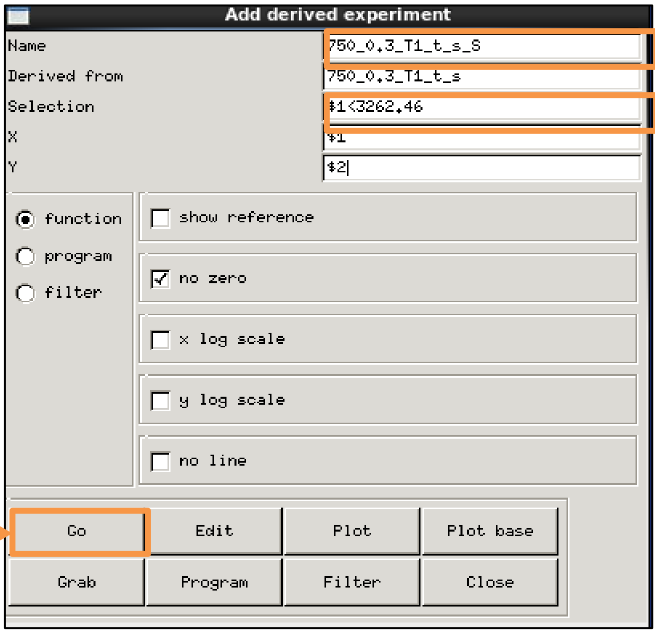
(3) Y axis: strain, second column

b. Derive an experiment

Once you add every curve for each experiment you want to observe, you need to filter those curves to determine the materials parameters E, C, D and R0.

i. Determine the Young modulus

First, we will determine the Young modulus by considering the beginning of the plots time_sigma and time_eto. We derive the time_sigma curve first and we are going to pick only the first linear part of the curve. Select the t_s curve you want to observe and click on Derive button, the next window is now open.

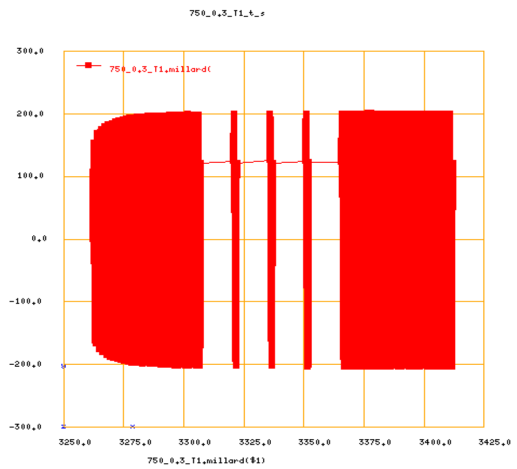


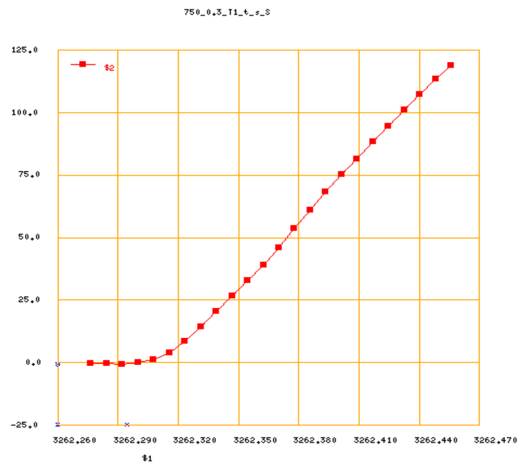
(1) Type a name, here we just add 'S' as start

(2) Filter the curve by adding a condition on time

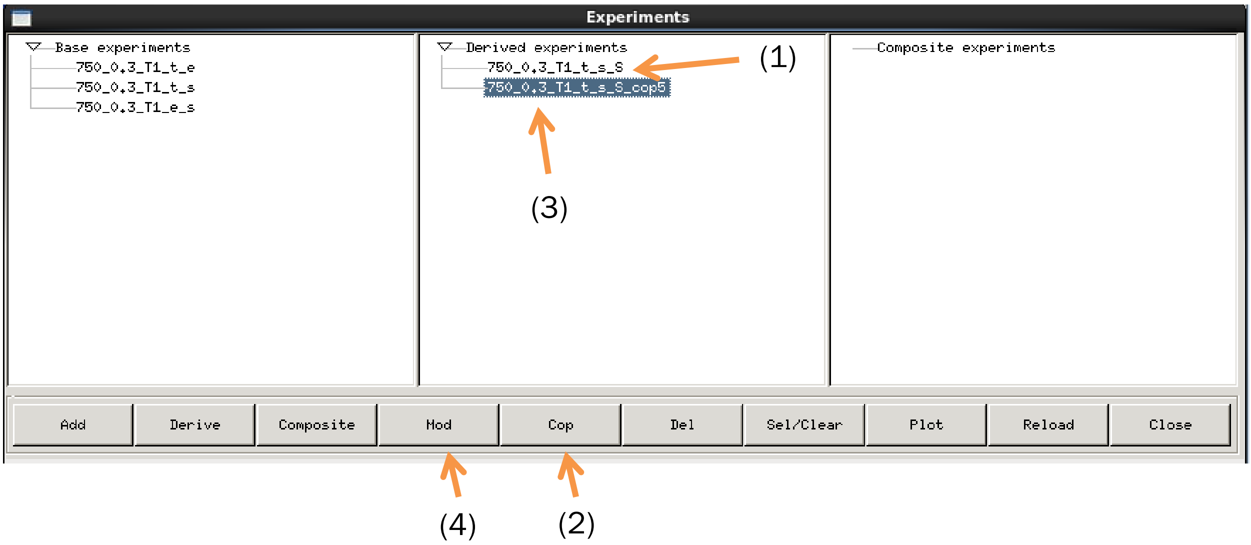
(3) Click Go to see the filtered curve and to confirm

See below the kind of curves you should have before and after this procedure:

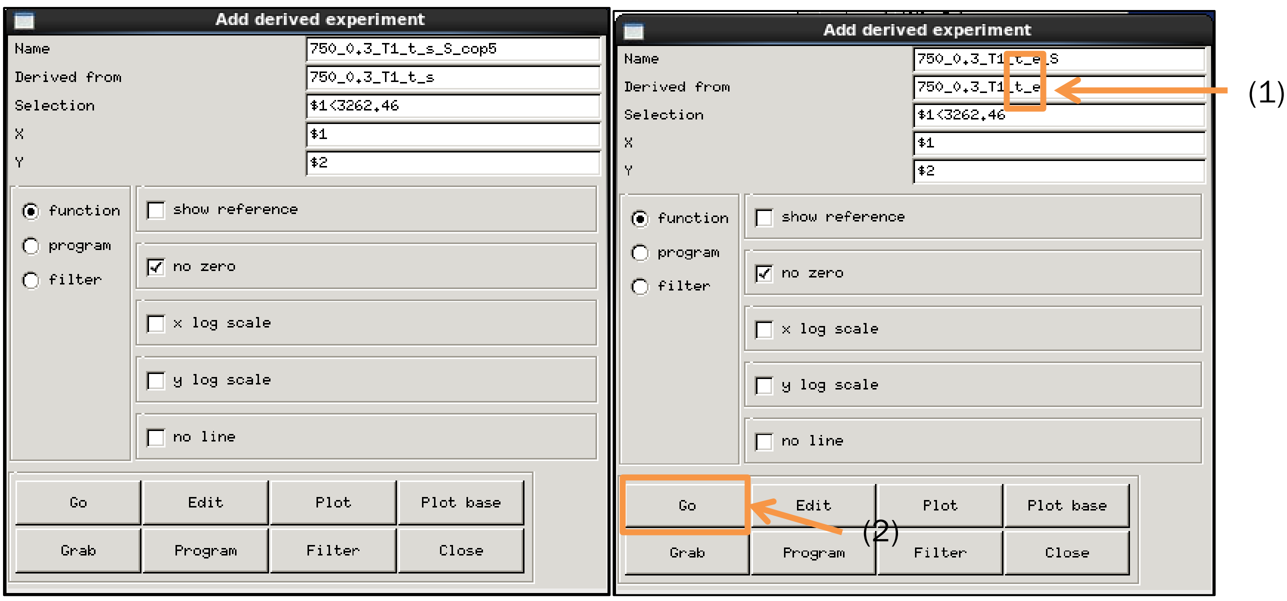




To derive the t_e curve, you need to apply the same filter condition than the one you use to filter the curve t_s. The easiest way to do that is to copy your derived t_s curve and to change the name of it and the initial curve as shown next:

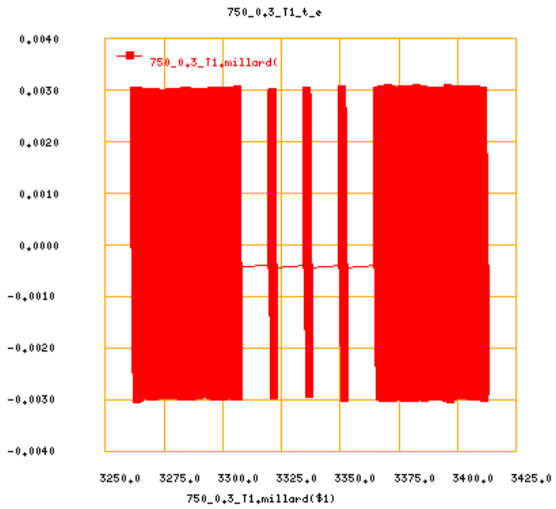


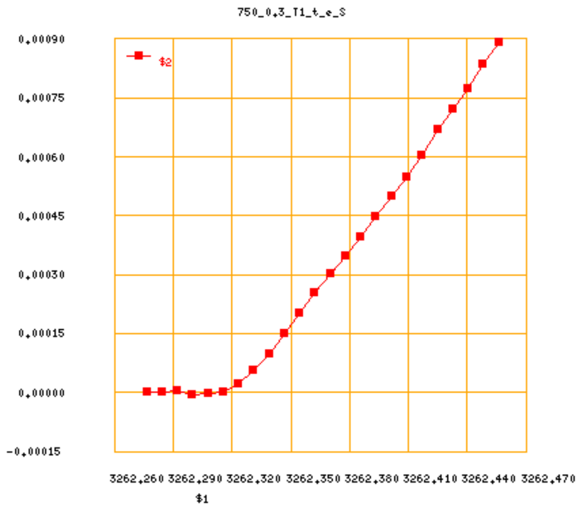
After this, a window will appear similar to the one we had to filter the t_s curve. You only



need to change the name of this new curve and the initial curve it will be derived from. The next picture will show you the window as it is initially and the changes we made.

These are the curves you should have before and after this action.



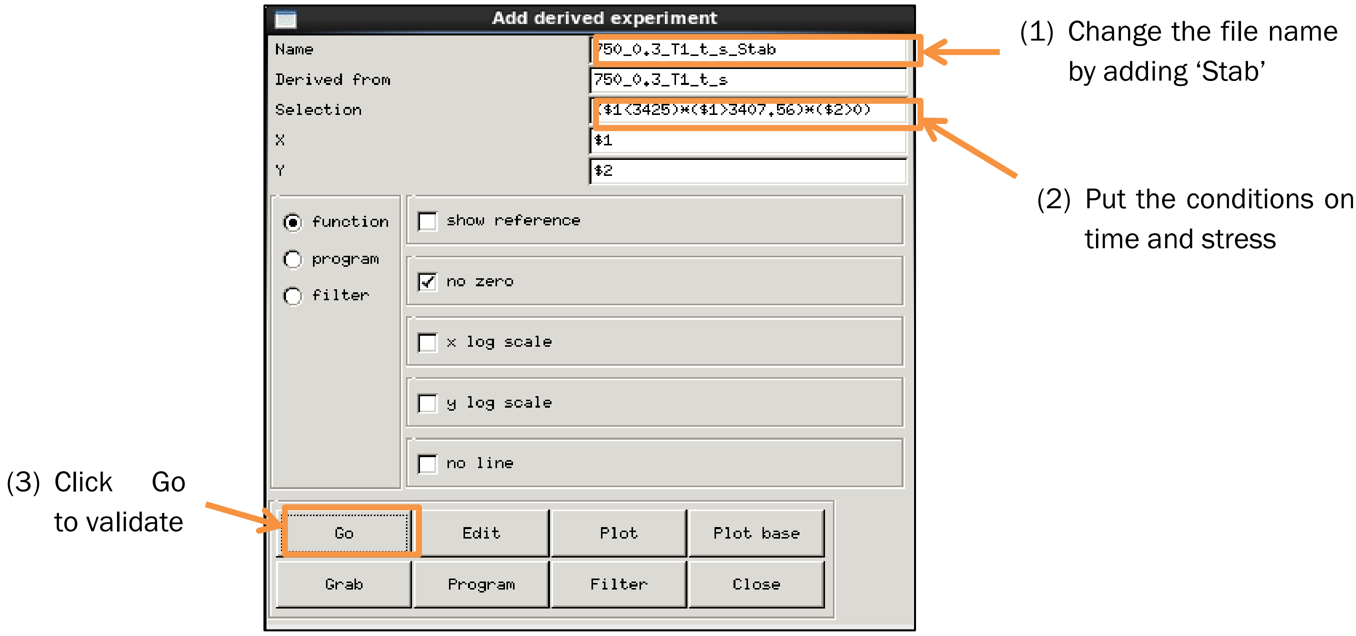


ii. Determine the C, D and R0 coefficients

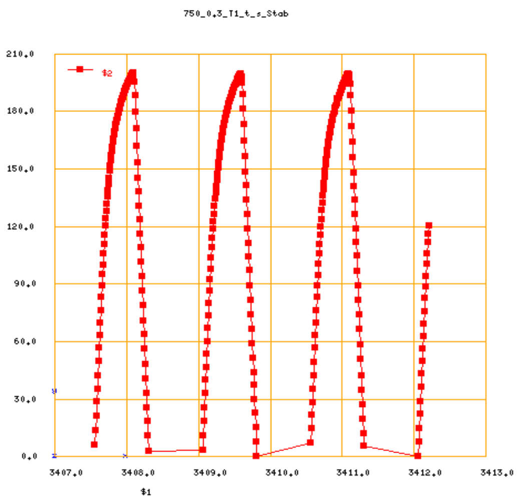
Next we will derive the same curves to identify the parameters R0, C and D. This time we will consider the last cycles of those curves, the last three cycles for the t_s curve and the last four cycles for the t_e curve.

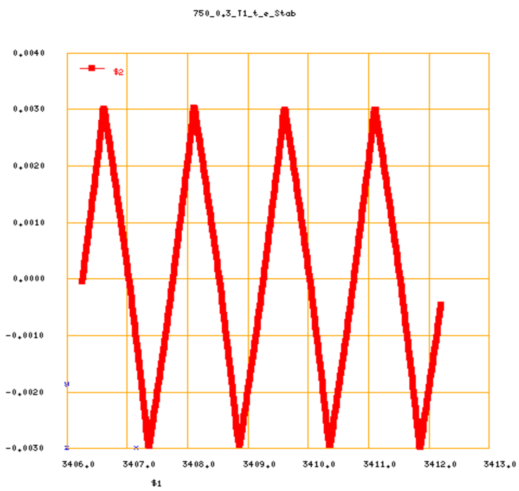
The last cycles are considered as stable, indeed we see that the material undergoes a hardening period at the beginning of the experiment and become stable by the end.

As seen previously, you have to filter those curves starting by the t_s curve and select the last three cycles by using conditions on time. Once the curve is filtered, we want to consider only the traction part of the experiment and this mean considering only the positive values of strain on the t_s curve. See the example below:



If you want to had multiple variables, you have to put each condition between brackets and to separate each condition by a star '*'. You have to do the same procedure to filter the t_e curve without considering only the positive part of the curve and by taking into account the last four cycles. These are the curves you should obtain by the end of these actions:





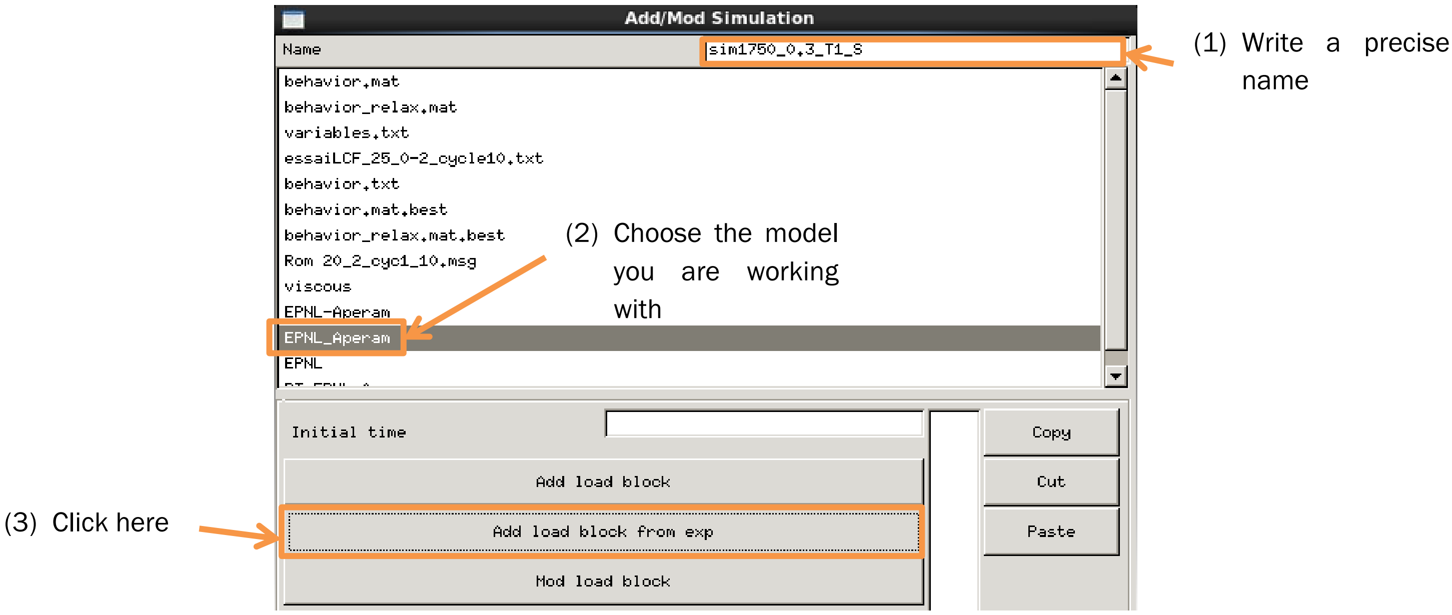
4. Simulation tool

The simulation tool gives us plots using material parameters we enter in the variables tool. It's a useful tool to compare the shape of the experimental curve and the curve we generate with specific parameters. This tool is using information from the variables tool and the experiments tool.

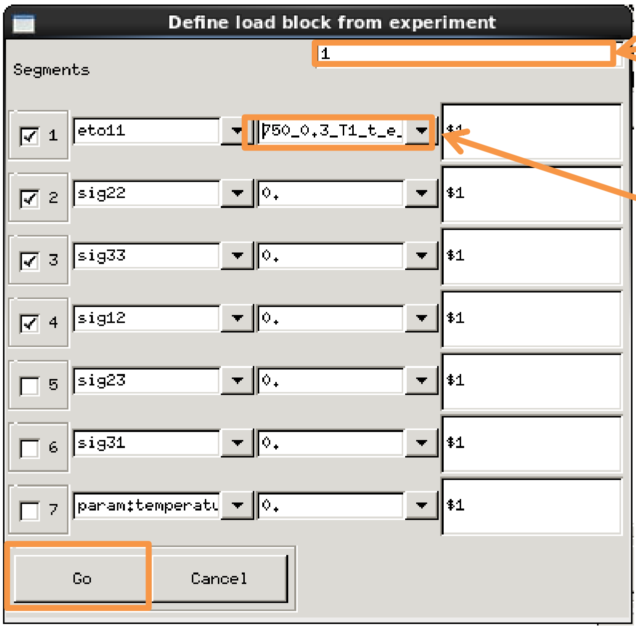
The simulation tool is presented like that:

We will simulate two types of behaviors resulting of the filtered curves from experiments called ‘Start’ and ‘Stable’.

Firstly, I will explain how to simulate a ‘Start’ curve. Press on the ‘Add sim’ button and a new window is now available. I will detail how to complete it, starting by the upper part of the window:



After you clicked on ‘Add load block from exp’, a new window is open and you will see next what to do here:

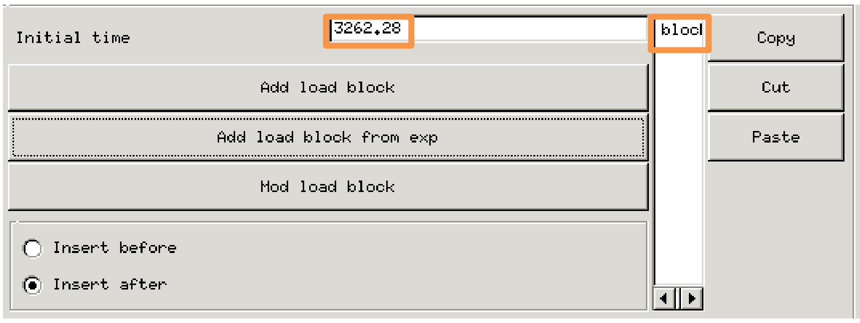


(1) Pick only one segment

(2) Pick the t_e_S experiment curve in the list

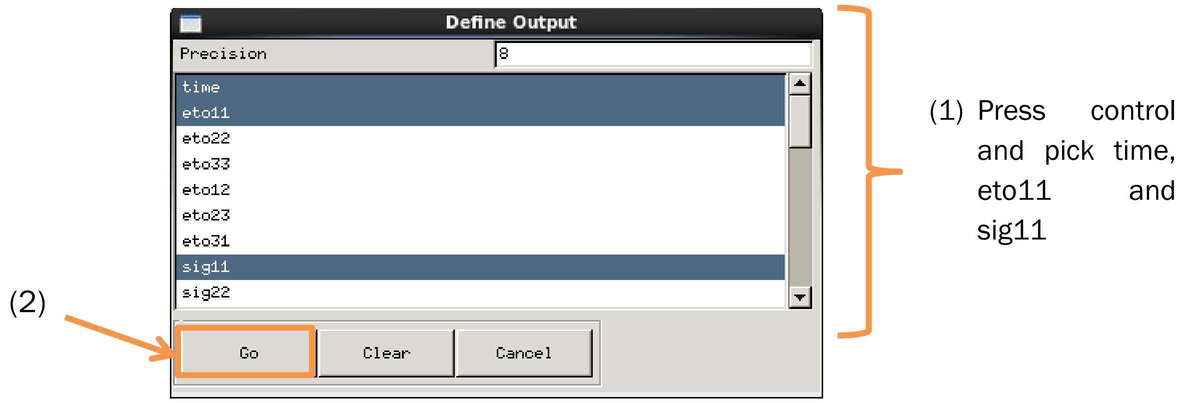
(3) Click Go

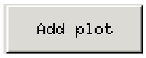
After you click Go, you can observe a change on the 'Add sim' window. The initial time is automatically written and a block is created.



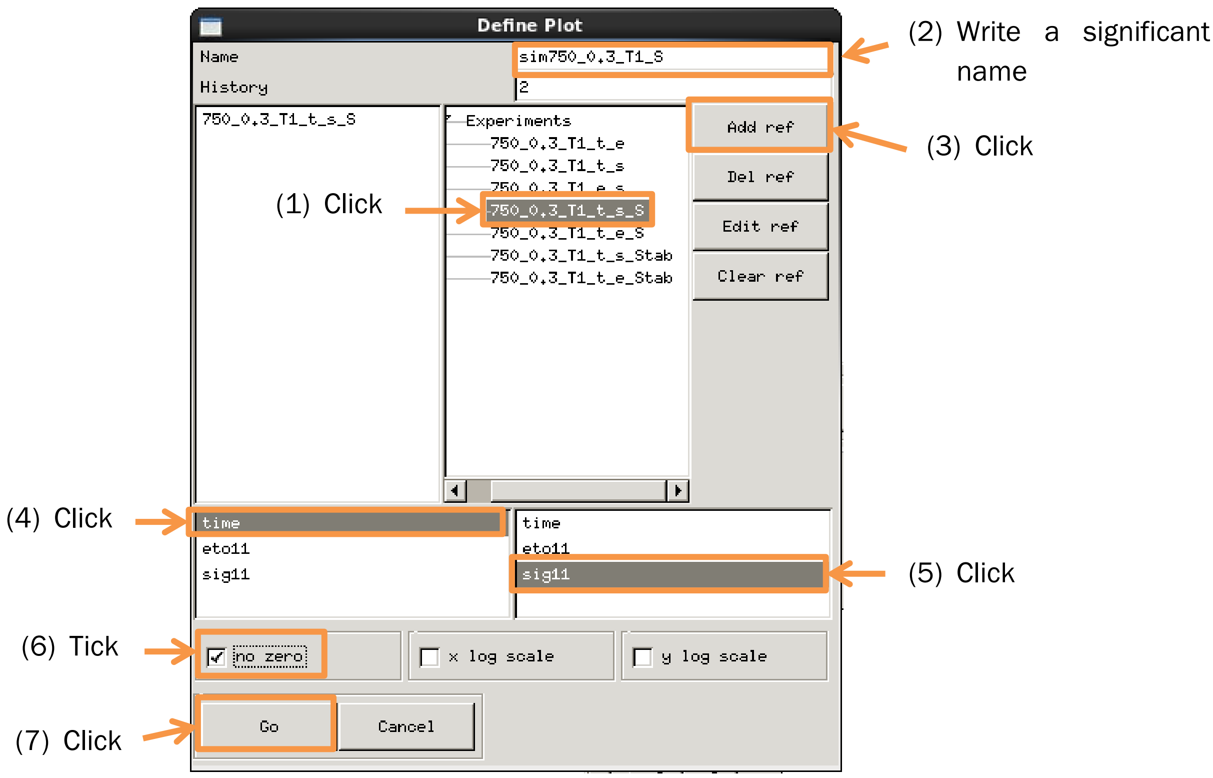
We will complete now the lower part of this window, starting with the 'Define output' tab:

When you click on this button, a small window appears with a list and you have to pick the three next output: time, eto11 and sig11.

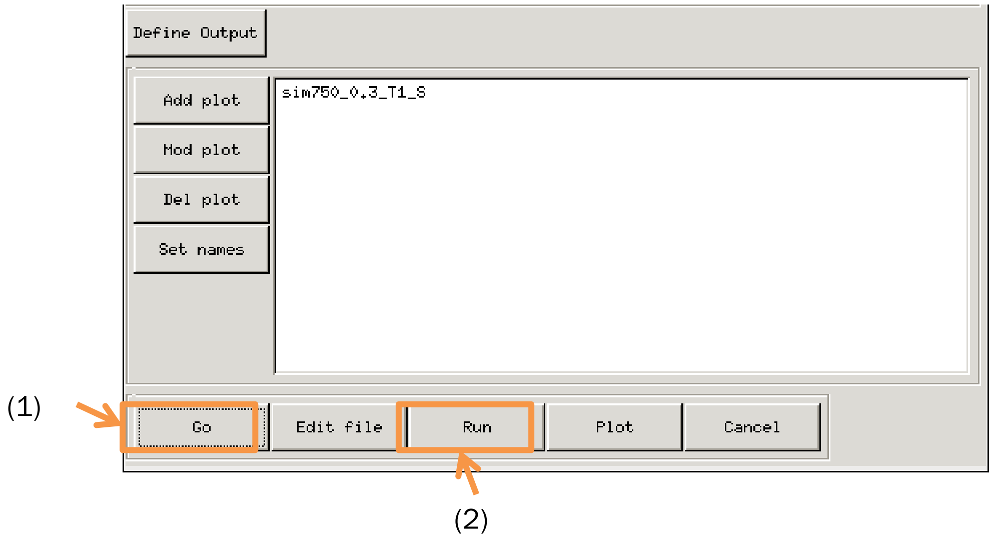


Next you have to add a plot to your simulation so you can compare the t_s curve from experiment and the t_s curve from simulation. Click on 'Add plot' tab 

and the next window appear and this is how it should be completed:



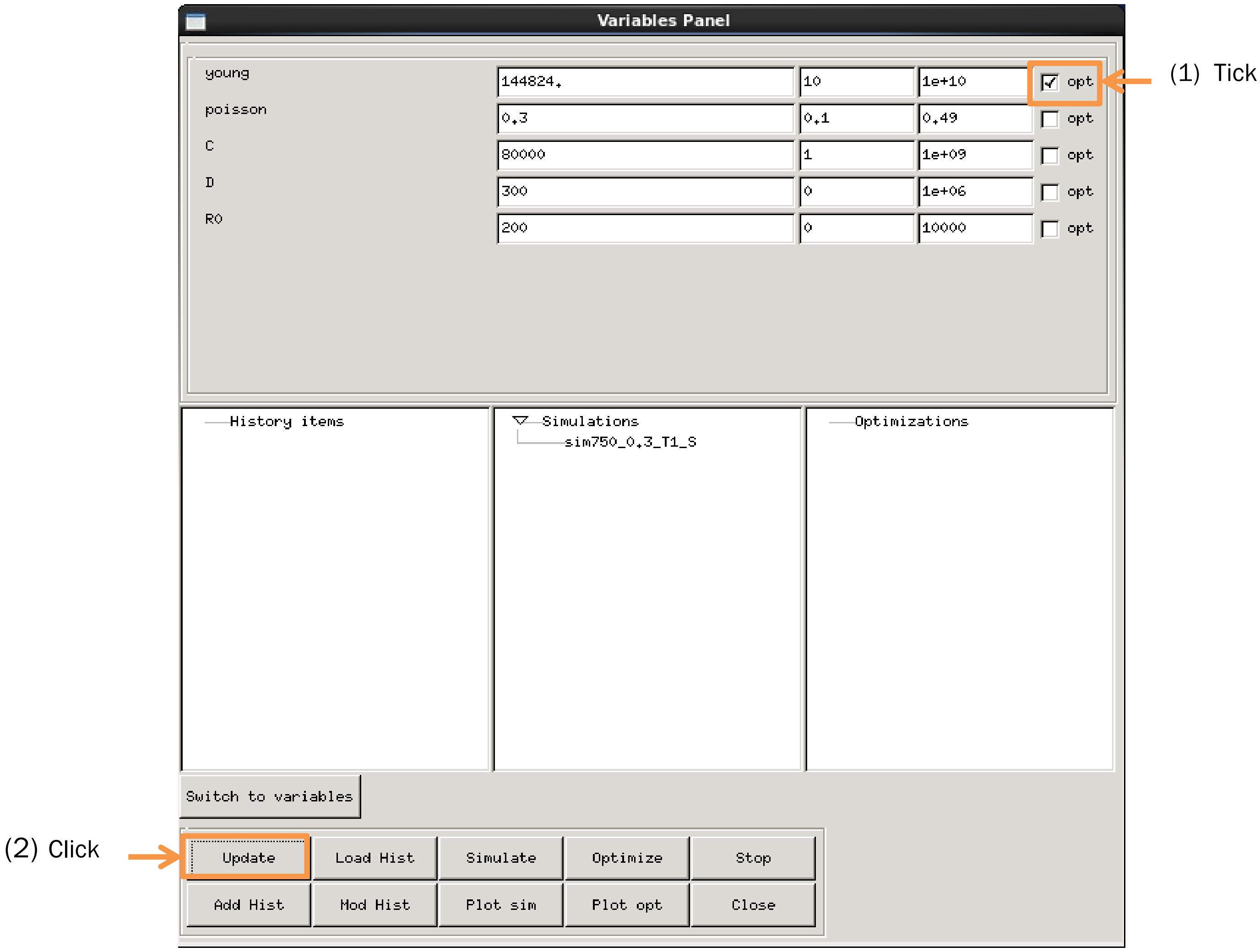
When the plot is added to the main window, your simulation is ready and you can click on 'Go' and then on 'Run' to see the result.



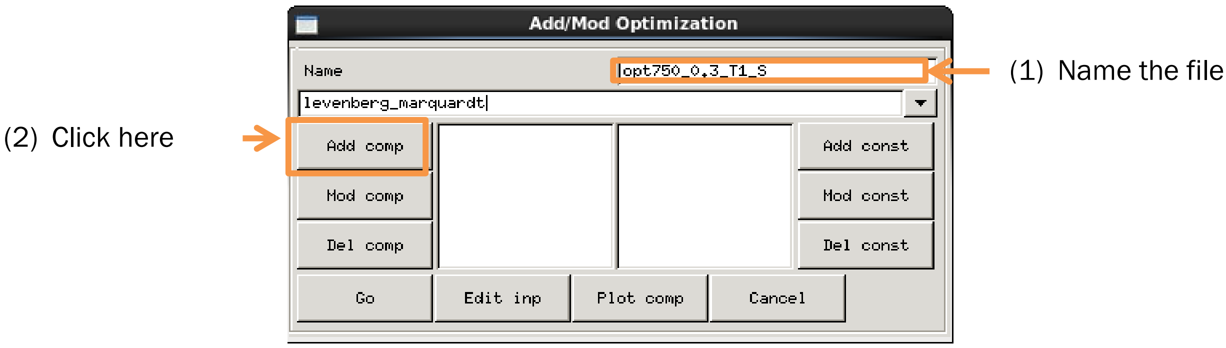
To simulate a ‘Stab’ curve, the procedure is the same but you need to be careful and to choose only ‘Stab’ curves instead of ‘S’ curves.

5. Optimization tool

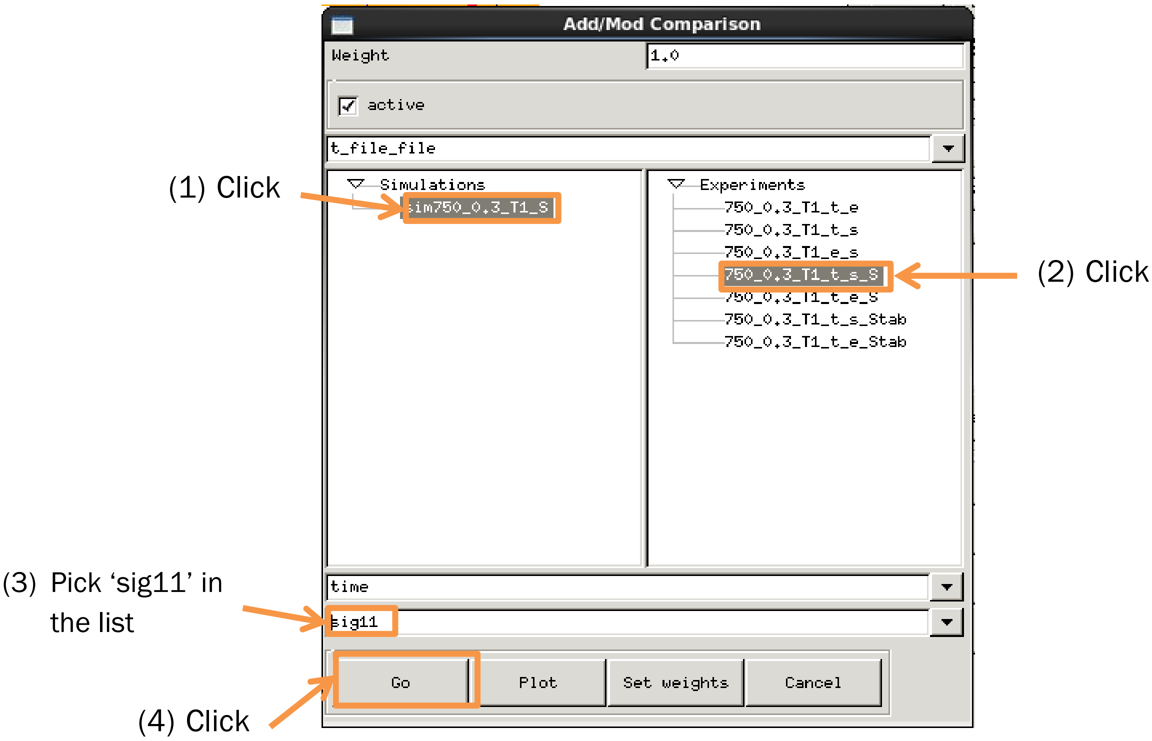
Optimization tool is use to optimize the material parameters in order to have a simulate curve close to the experiment curve. This is how this software is giving us the parameters we want (E, R0, C and D). There is also here two types of optimization, one type to determine the best value of Young’s modulus and a second one to determine the best values of R0, C and D. The first one uses the ‘S’ curves and the second uses the ‘Stab’ curves. First you have to tick the parameters you want to optimize in the ‘Variables panel’ window (see chapter 2.2) and update the data.



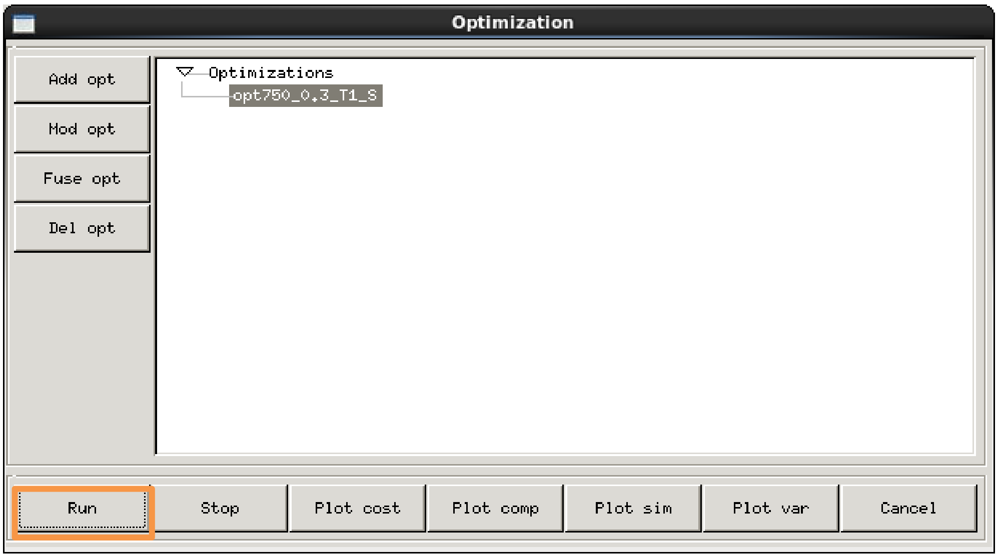
I will detail now the procedure to optimize the Young's modulus with the optimization tool. You need to click on 'Add opt' tab to open the related window that looks like this:



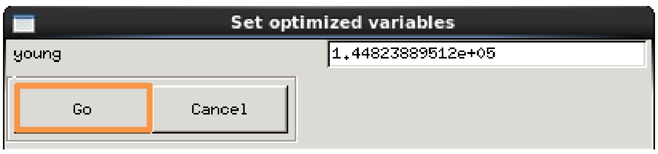
We need to specify the components we want to optimize; this is why we need to click on the 'Add comp' tab and to fill the window as shown next:



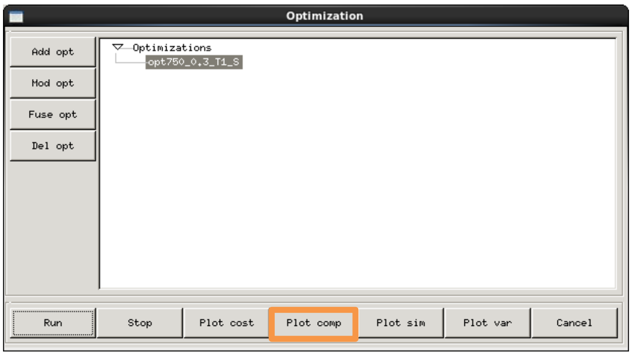
This component is now present in the 'Add opt' section, your simulation is ready. Click Go in this section and you can run this optimization.

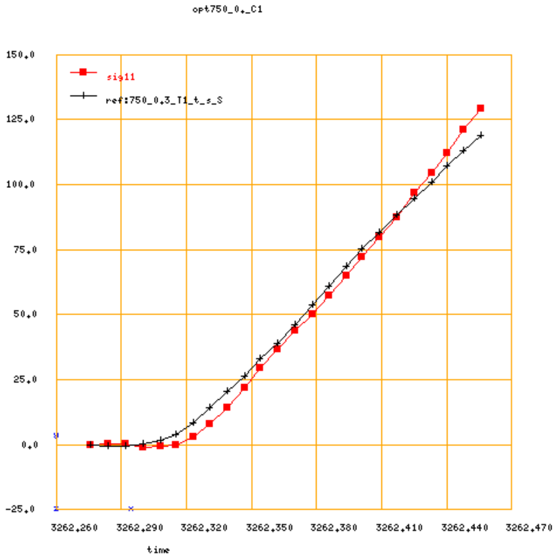


The result calculated by the computer will be shown on a small window where you can click Go to use this result in the variables panel.



If you want to see how the new component is compared to the experiment you have to click on Plot comp and you will see the experimental curve and the simulated curve with the new value of Young's modulus plot together. If you see that the optimized curve is not following well the simulated one by the end, you need to choose a bigger value of R0 in the variables panel and run the optimization again to find a new value of E that will fit better the reference curve.



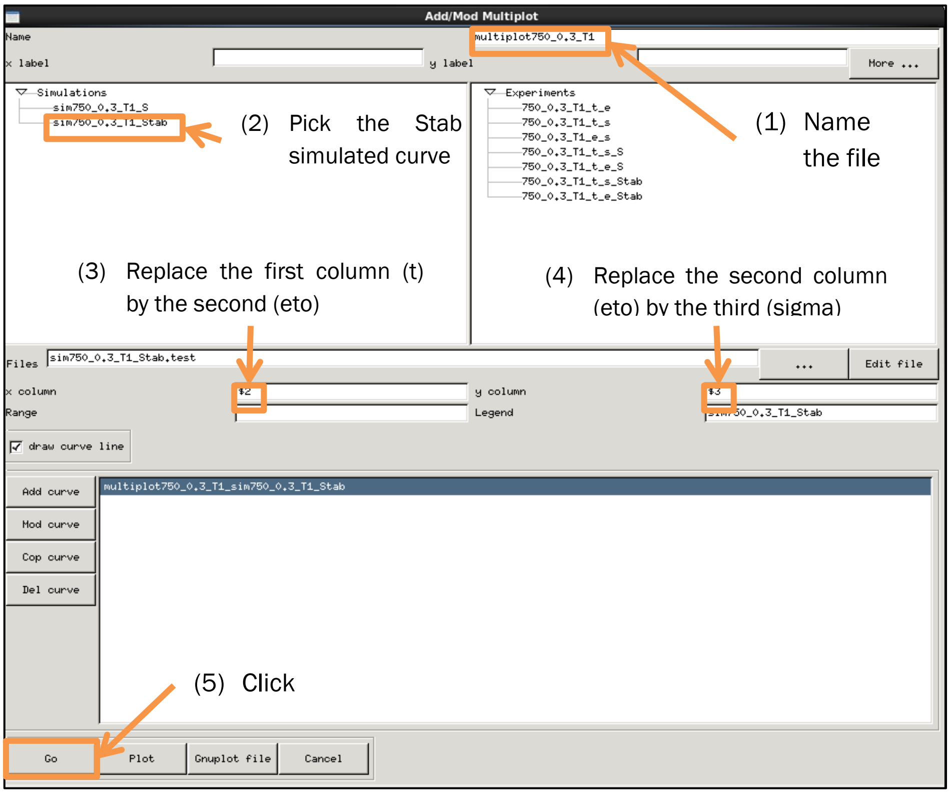


If you see that curves are crossing each other, you can also modify the Young's modulus manually in the variables panel until the curves look parallel following the straight part. Usually the new value is lower than the optimize one.

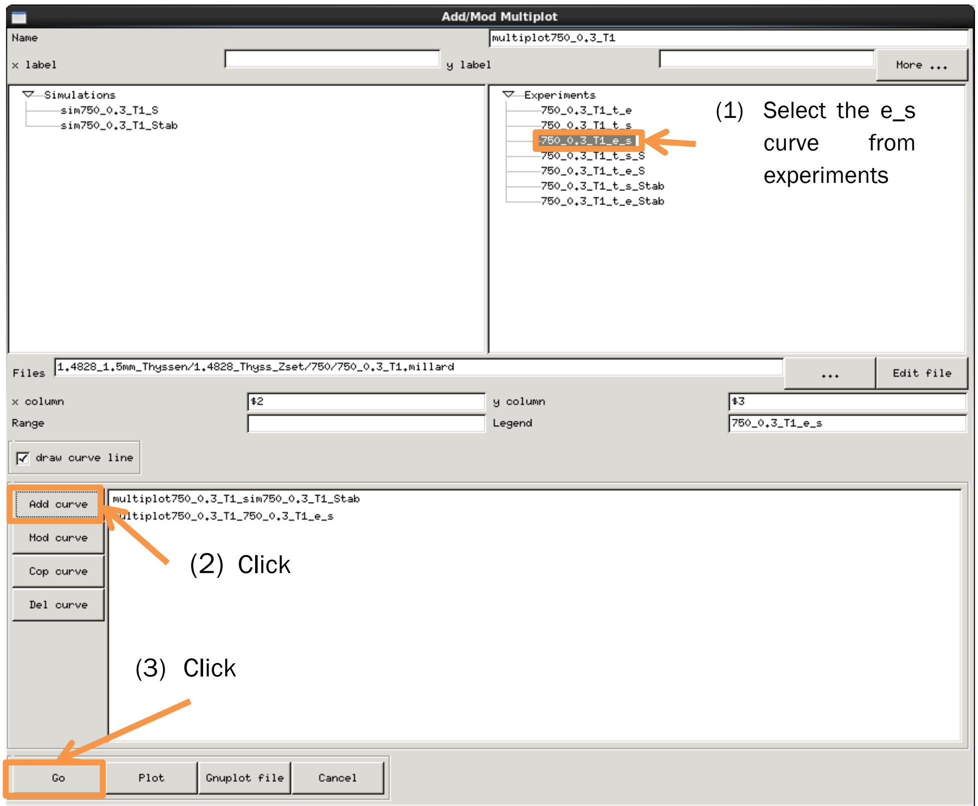
The next step is to optimize the parameters C, D and R0 by using the same procedure and carefully choose only 'Stab' curves when you define the component and change parameters to optimize on variable panel before running the optimization, you need to unpick the Young's modulus.

6. Plot tool

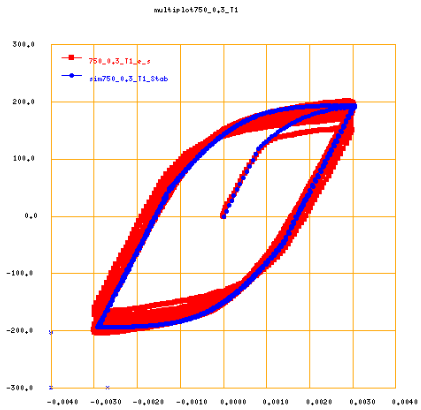
When you have every parameter with optimize value, it is time to compare your model with the experimental model thanks to the plot tool. This tool gives us a chance to superimpose the experimental e_s curve with the simulated one. You first need to click 'Add multiplot' and the next picture will show you how to add the first curve from the simulation:



Then you need to add the e_s curve from experiments by following the next procedure:



We only have two curves to compare when we are working with single-step test; next picture show the result we have for our example:



III. HOW TO USE Z-SET TO IDENTIFY MATERIAL PARAMETERS FROM A MULTIPLE-STEP TEST

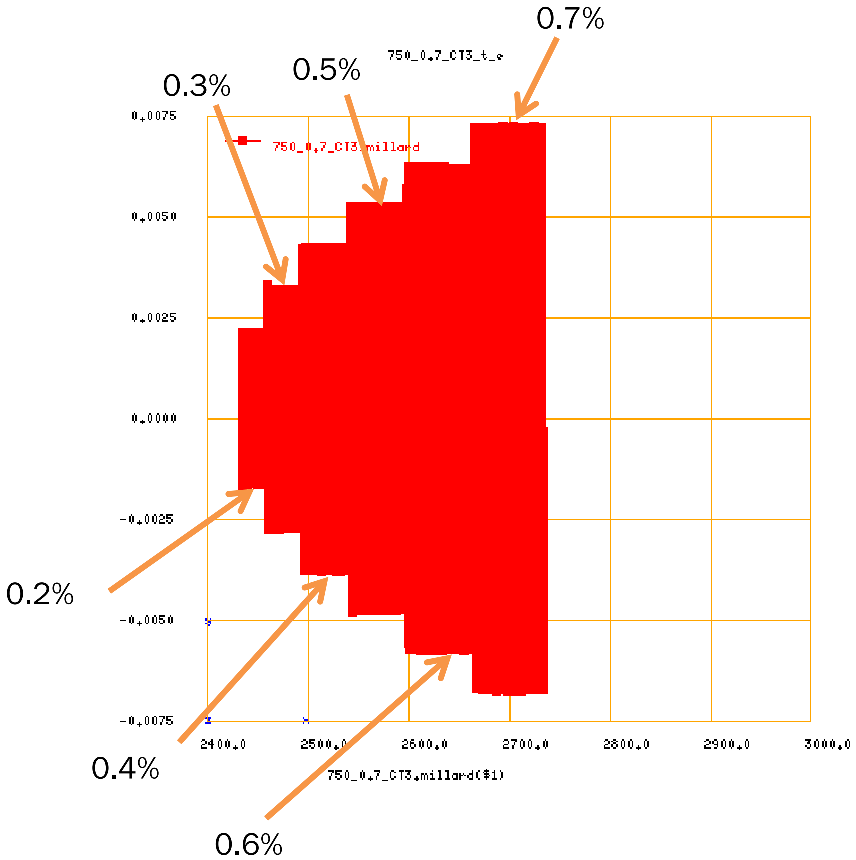
If you have to analyze a multiple-step test the procedure is lightly different from the single-step test procedure. Indeed, in a single-step test we only consider one strain rate at a time while in a multiple-step test the strain rate is increase after a fixed cycle's number (see following document here).The previous procedure will be different from the 'Experiment tool'.

1. EXPERIMENT TOOL

As seen before, this tool makes plots from our experimental results and we can filter those plots to determine the following material parameters: E, R0, C and D.

The first step of the procedure is the same than before, you need to plot three plots from your experimental results time_eto, time_sigma and eto_sigma. From this, you have to filter each strain rate from the t_s and t_e curves like before by considering the first cycle as start ('S') and the last cycles as stable ('Stab').

This is what a t_e multiple-step test curve looks like with the different strain rate to consider in this case.



Strain rates are the same for t_s curve. Like this, in the Experiment tool, you have only three base experiments and 24 derived experiments. There are two types of derived experiment, the S type to determine the Young's modulus E and the Stab type to determine the R0, C and D coefficients, apply to both t-e and t_s curves for each applied strain rate.

To filter those curves, you need to follow the same procedure than before.

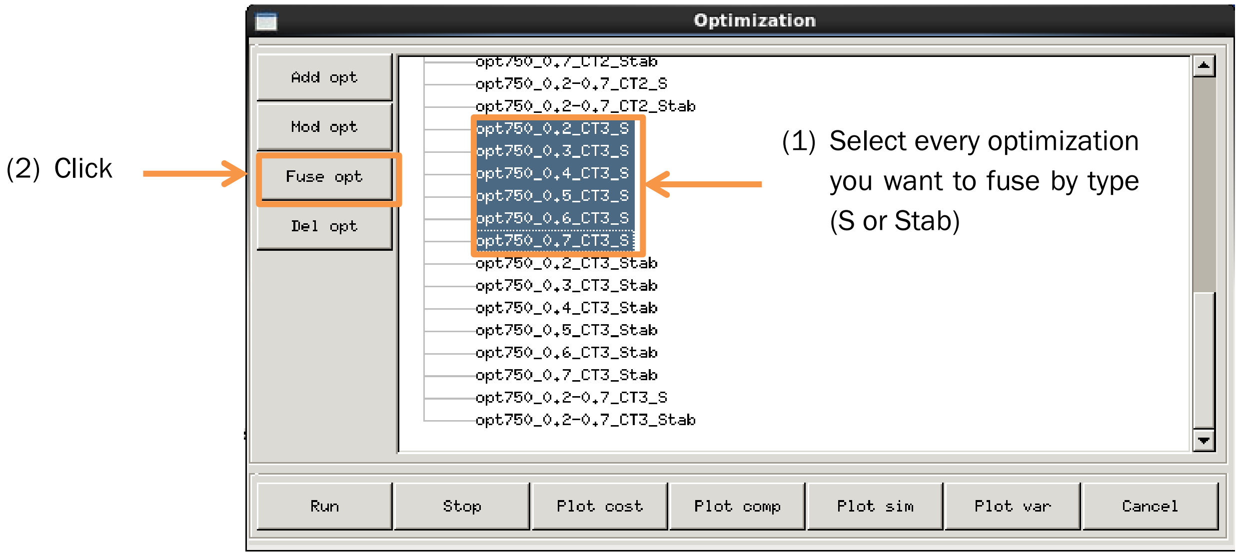
2. Simulation tool

As for a single-step test, each derived experiment need to be used in the simulation tool. That means that we will have two simulations (S and Stab) for each strain rate. Except for this, the procedure is the same than the one showed earlier.

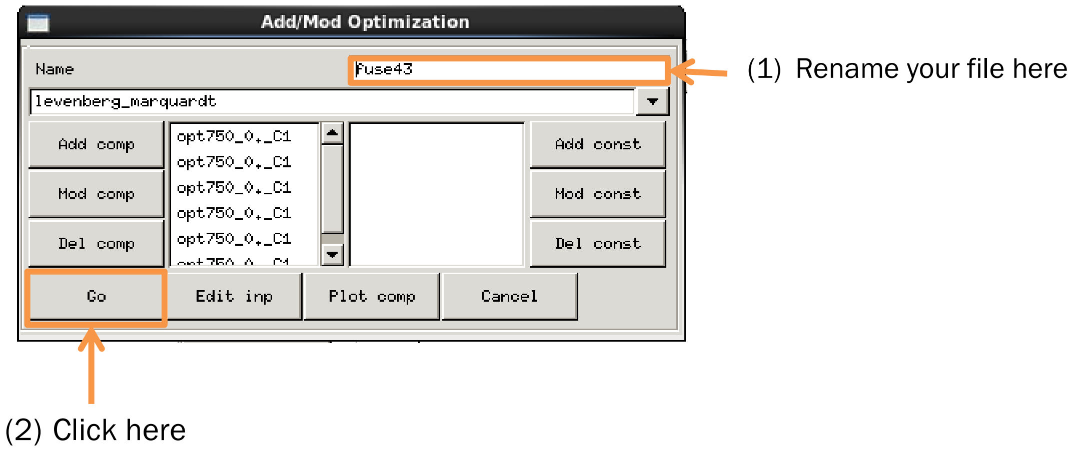
3. Optimization tool

This tool is used almost the same way for multiple-steps and for single-step tests (see procedure here).

We are creating an optimization for each simulation, and once this step is done we can fuse all those optimizations into a single one in order to have only one value for each parameter that optimizes every simulation curve. The next picture will show how to do this.



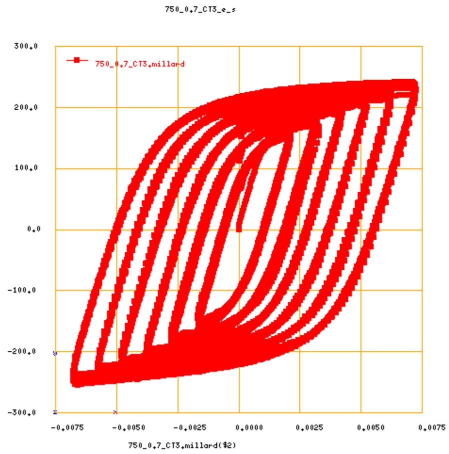
When you click on ‘Fuse opt’ button a new element will appear in the list call “fuse…” you can rename it if you select it and then click on ‘Mod opt’.



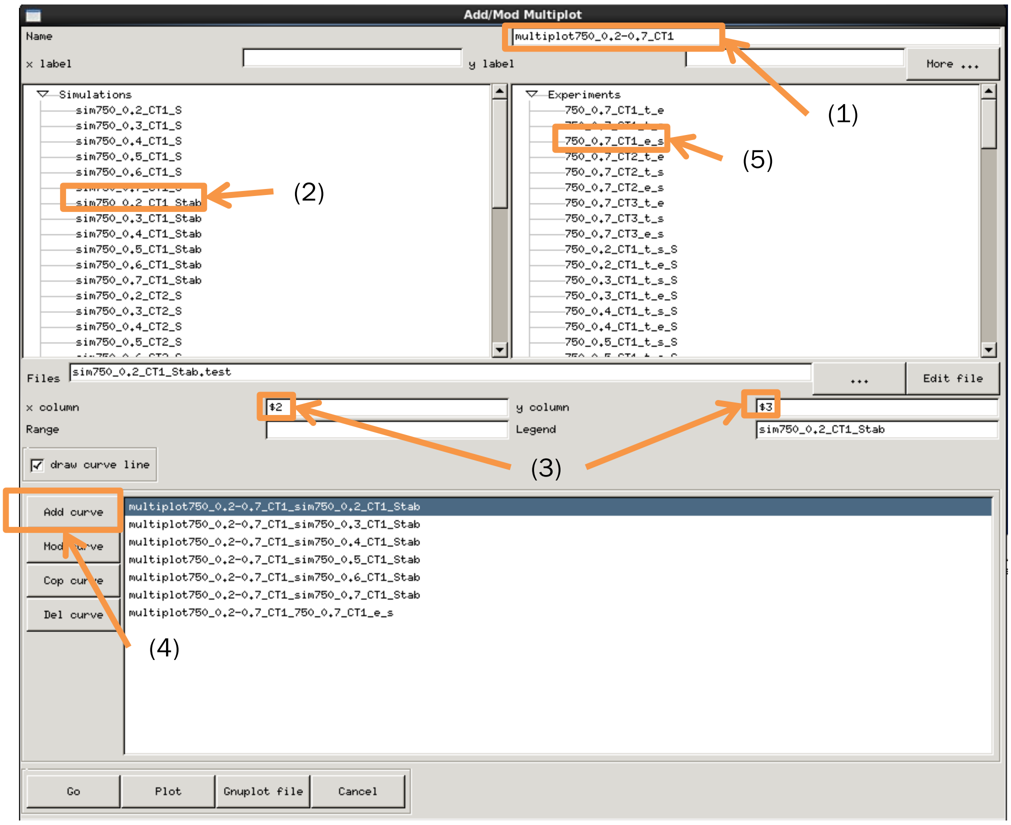
You can now use those fused optimization to determine the material parameters like before by choosing the S optimization to compare the Young’s modulus you found at first and this new value and the Stab optimization to find the R0, C and D coefficients.

4. Plot tool

This tool will be a bit different to use than before, indeed the initial e-s plot we have is composed by the addition of the e_s curve for each strain rate. This is what kind of curve we initially have:



In the plot tool we want to draw our simulated curves and the experiment curve on the same plot to compare their shapes. To do so we will need to add multiple simulated curves for only one experimental curve in the plot tool as shown next:



The first thing to do when you see this window is to correctly name the file you create. Then you need to select the first Stab simulation and to select the second column for x value and the third column for y value, this will give you the e_s curve of this simulation. Then click 'Add curve'. You have to repeat this action for each Stab simulation and then the last curve to add to the plot is the experimental one from the right list as highlighted on the fifth point of the picture. Then click 'Go' to plot those curves. This is what we have with our example: